it may be claimed with full justification that in the long run, we owe our entire existence on earth to such processes. For, firstly radioactivity plays an extremely important part in determining the temperature and climate of the earth's surface, and secondly, it is owing to such reactions in the sun that it shines on the earth and sustains life on our planet.

Nuclear reactions on a large scale are in fact taking place in the interiors of the stars. We know today that the energy which the stars—including our own sun—radiate into space is the product of such nuclear reactions. The source of this radiated energy was a puzzle for a long time, the solution of which called for a great deal of mental skill. We know that our sun has shone upon the earth with approximately the same intensity for at least two thousand million years, and in former years nobody could comprehend why it had not expended its total energy a long time ago. The solution of this problem had to wait for nuclear physics, and today we are in the position to specify exactly the only process that can possibly account for it. This solution was established as a result of three studies carried out by Atkinson and Houtermans, v. Weizsäcker, and Bethe. We cannot describe here the way that led to this solution; we can merely quote the result. It is a question of a certain sequence of formulae, which may be stated here, as follows:

(1) ${}_6C^{12} + {}_1H^1 \rightarrow {}_7N^{13}$

(2) ${}_7N^{13} \rightarrow {}_6C^{13} + {}_1e^0$

(3) ${}_6C^{13} + {}_1H^1 \rightarrow {}_7N^{14}$

(4) ${}_7N^{14} + {}_1H^1 \rightarrow {}_8O^{15}$

(5) ${}_8O^{15} \rightarrow {}_7N^{15} + {}_1e^0$

(6) ${}_7N^{15} + {}_1H^1 \rightarrow {}_6C^{12} + {}_2He^4$

This series consists partly of proton-induced reactions (${}_1H^1$ is the symbol of the proton), partly of transmutations by the emission of a positron (${}_1e^0$). The initial substances are carbon of mass number 12, and hydrogen. It is a known fact that the stars consist to a large extent of hydrogen, and that carbon also is present in them in small quantities. The hydrogen is already in the form of protons, since as a result of the enormous velocities due to the high temperatures (10–20 million degrees C.) prevalent in the interiors of stars, practically all the hydrogen atoms have been stripped of their single planetary electrons. This enormous velocity enables the protons to penetrate the other nuclei.

So, first, a nitrogen nucleus is formed from an ordinary carbon nucleus and one proton (1). This nitrogen nucleus is unstable and changes by the emission of a positron, into a carbon nucleus (2). This newly produced carbon nucleus is a heavy isotope of that which started the process. (The entire process is well